DEEPAK KATUKURI

AI-NATIVE PRODUCT BUILDER · FULL-STACK SHIPPING · AI AGENT SYSTEMS



CONTACT

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TECHNICAL SKILLS

Product & AI Agent Tooling: Claude / Claude Code, structured prompt & context engineering (persistent instruction files), PRD-style spec writing for AI-agent execution, ATS/keyword-driven document QA

Full-Stack: React, Vite, Tailwind CSS, React Query, PostgreSQL (Supabase-ready), FastAPI, Recharts

Automation & Data: Python, SQL, PL/SQL, LangChain, workflow automation, formula-driven Excel tracking, Git/GitHub

Collaboration: Agile/Scrum (sprint planning, stand-ups), stakeholder-requirement translation, QA/UAT

ML/DL (secondary): PyTorch, hybrid retrieval (BM25 + dense, RRF), pgvector, scikit-learn

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

LANGUAGES

English – Fluent

PROFILE

AI-native product builder and Data Science MSc candidate who ships end-to-end: full-stack products, structured specs, and the AI-agent tooling that runs day-to-day work. Designed a 5-role AI agent pipeline that turns a raw brief into a verified, tailored deliverable through persistent context files and automated review — the same discipline of clear requirements, fast tool pick-up, and removing manual work that colayer's product team runs on. Also shipped a full-stack health app (React + FastAPI + PostgreSQL) solo, from feature scoping to UI to backend, and built a hand-rolled RAG retriever validated against a LangChain rebuild — backed by 2.5 years of production software engineering. Open to working-student / part-time arrangement alongside studies.

PROJECTS

AI Agent Ops System – Multi-Role Instruction Pipeline for Application Automation

Claude (Agent SDK) · Persistent context engineering · Python · Excel automation

- Designed a 5-role agent pipeline (diagnosis → rewrite → review → recruiter-simulation → hiring-manager-simulation) driven by a persistent instruction file, turning a raw spec into a verified, tailored deliverable in one structured pass instead of manual drafting.
- Built cross-session project memory with fact-verification guardrails – dated correction logs and exact date/number checks – that caught and fixed 3+ factual errors (a wrong duration claim, a stale project framing, an overstated fix status) before they reached a submitted document.
- Automated ATS keyword scoring and a formula-driven tracker, cutting per-application tailoring effort while keeping every claim traceable to a verified source – zero fabricated content across output.

RA Nutrition & Lifestyle App – Full-Stack Product, Built Solo

React · FastAPI · PostgreSQL · Recharts · personal product build

- Owned the product end-to-end – scoped features, designed the data model (4 logical schemas), and shipped a React/Vite/Tailwind frontend over a FastAPI + PostgreSQL backend with JWT auth and token refresh.
- Built a rule-based recommendation engine and a weekly analytics dashboard (Recharts) that closes the loop from raw meal/symptom logs to actionable output for the end user.
- Scoped a V3 roadmap (ML forecasting tier: scikit-learn/XGBoost, PyTorch LSTM/GRU/TCN for flare-risk prediction) before building it – prioritizing what to ship next over building everything at once.

RAG Pipeline – Hand-Rolled Retrieval, Validated Against LangChain at Scale

Python · LangChain · BM25 + dense hybrid · pgvector · Ollama

- Built a hand-rolled hybrid retriever (BM25 + dense bi-encoder, RRF fusion, cross-encoder reranking) across 4 iterative stages, reaching 0.833 recall@5 / 0.672 MRR on a 55-question labeled golden set – a metrics baseline before adopting any framework.
- Rebuilt the same retriever on LangChain over the identical corpus and labels, caught a silent 0.06-recall regression from a framework default, and fixed it to 0.854 recall@5 – evidence over blind trust in a tool.
- Extended the pipeline into an agentic, self-correcting retrieval loop that evaluates its own retrieval confidence and re-queries/reformulates before generating, instead of a fixed single-pass retrieve-then-generate.
- Scaled the corpus ~100x (7 to ~740 documents) with a resumable ingestion pipeline feeding a pgvector-backed store.

EXPERIENCE

Software Engineer — Mar 2022 – Sep 2024

ValueMomentum

- Translated business requirements from underwriting and claims stakeholders into technical specs during sprint planning, then accelerated releases by running QA/UAT test cases and supporting CI/CD deployment across the full policy lifecycle.
- Migrated full policy, claims, premium, and customer history for the Surety Bonds line from a legacy policy admin system into OneShield PAS, building and validating PL/SQL migration scripts across customer, policy (PID), claims, and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty, and kept records current with a weekly backfill script pending a permanent fix.

ML Intern — Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble and deployed it via IBM Watson ML, communicating results to non-technical reviewers with Matplotlib/Seaborn visualizations.